

THE 30 DOLLAR PUMP THAT PUSHES WATER UPHILL

No fuel, no power, no bill. Parts list, ratio math, and sizing rules for a hydraulic ram.

A ram runs on water hammer, the bang in the wall when a tap slams shut. Creek water races down a rigid pipe, a valve slams it to a dead stop, and the pressure spike shoves a slug of water past a one way valve into an air chamber. The squeezed air pushes it up the hill. Then the waste valve drops open and the cycle repeats, 40 to 90 times a minute. A well tuned ram often settles near 60, a resting heart rate.

STEP 1: MEASURE BEFORE YOU BUY

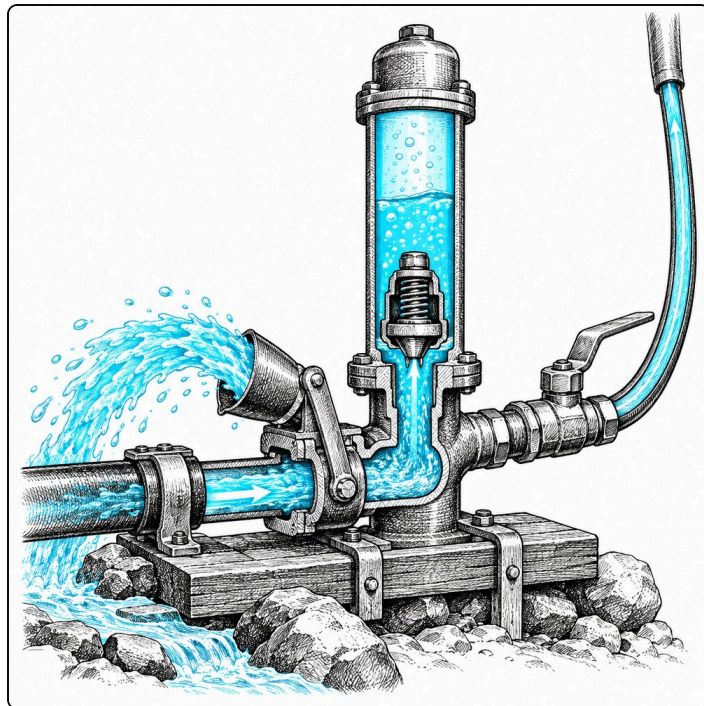
- 1 Flow.** Hold a 5 gallon bucket under the flow where the intake will sit and time it. 30 seconds means 10 gallons a minute. Two minutes means two and a half, below the practical floor.
- 2 Fall.** Measure the drop from the intake to where the pump will sit.
- 3 Lift.** Measure the height from the pump up to your tank.

STEP 2: THE MATH

Formula	Flow times fall, times 0.6, divided by lift = gallons a minute delivered uphill.
Example	10 gallons a minute, 5 feet of fall, 50 feet of lift: $10 \times 5 = 50$, $\times 0.6 = 30$, divided by 50 = 0.6 gallons a minute.
860 a day	0.6 gallons a minute, around the clock, is about 860 gallons a day, roughly three thousand two hundred liters.
1 gallon	The federal emergency guideline per person per day for drinking.
3 gal/min	The practical minimum flow, with a few feet of fall. A spring line or stock pond overflow can do it.
22 psi	Lifting water 50 feet takes 22 pounds per square inch. The slam in plastic pipe carries roughly 40 to 80.

STEP 3: SIZING RULES

- ☐ **Drive pipe rigid and long**, about 5 to 10 times the vertical fall. Too short and the slug is too light to hammer.
- ☐ **Fall to lift sets the yield.** Lift 6 times your fall and you keep about 1 gallon in 10. Lift 20 times your fall and you keep a trickle.
- ☐ **Keep the chamber mostly air.** Every few weeks, drain it and let it refill with air, and the beat comes back.



Waste valve hanging open, delivery valve pointing up into a capped air chamber.

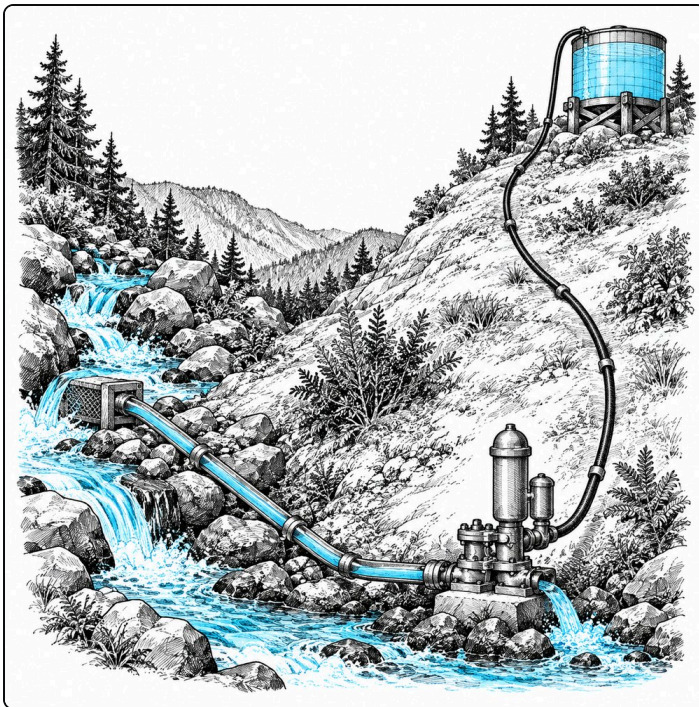
PARTS LIST, ABOUT 30 DOLLARS

- ☐ Rigid drive pipe from the creek intake down to the pump
- ☐ Screen on the intake to keep leaves and grit off the flap
- ☐ A tee at the bottom
- ☐ Brass swing check valve, flap hanging open downward: the waste valve
- ☐ Spring check valve pointing upward: the delivery valve
- ☐ Capped length of 4 inch pressure rated plastic pipe: the air chamber
- ☐ Ball valve on the delivery line to throttle it
- ☐ A couple of unions so you can take it apart
- ☐ Ordinary black poly pipe for the delivery line, hundreds of feet if needed

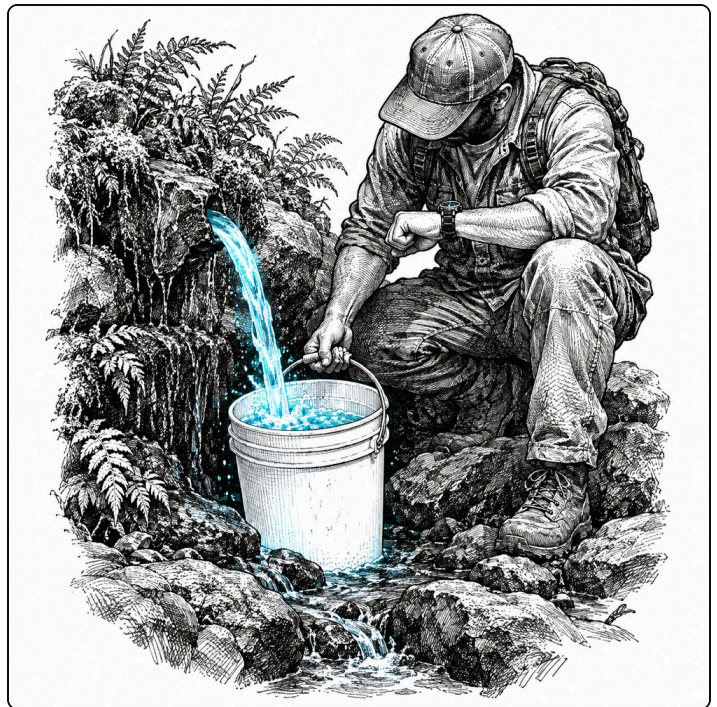
That is the common one and a quarter inch design in plastic fittings with basic valves. Brass valves push it higher, and they are the one place worth spending, because that flap gets slammed 60 times a minute for years.

TUNING

A heavier waste valve, or one that travels farther before it shuts, means fewer beats and bigger slugs. Lighter means more beats and smaller ones. Adjust until it settles into a steady clack you can hear from the porch.



Creek, rigid drive pipe down to the ram, delivery line up the hill to the tank.



Time a 5 gallon bucket at the intake to measure your flow.

THE LIMITS AND THE CAUTIONS

- ☐ No moving water with a drop, no ram. It cannot lift from a still pond or a well.
- ☐ No flexible hose for the drive pipe. It swells on the slam and swallows the spike. No rigid pipe, no hammer. No air, no spring.
- ☐ Pressure rated pipe only for the chamber and valve body. Never thin wall drain pipe.
- ☐ Freezing splits a standing drive pipe. A winter ram keeps running or gets buried below the frost line.
- ☐ Creek water is not drinking water yet. Filter it before anyone drinks it.

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